

Sofar $10 + 30 + 10 = 50$

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CMI
BVR

Analysis I/B1
Quiz 2

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12:30 - 13:10

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Can use anything done in class, state it and use.

Q1: Consider the curve: $f(x) = x^{25}$ on $[0, \infty)$. At what point $(u, f(u))$ is the tangent to the curve parallel to the chord joining $(0, 0)$ and $(1, 1)$?

[3]

Your answer with justification

→ Slope of tangent to a curve at x is given by $f'(x)$

$$\text{Here, } f'(x) = 25x^{24}$$

→ Slope of line connecting $(0, 0)$ and $(1, 1) = 1$ ($\frac{1-0}{1-0} = 1$)

→ Since the tangent is parallel to the chord, the slopes should match.

$$\therefore 25x^{24} = 1 \rightarrow x^{24} = \frac{1}{25} \rightarrow x = \sqrt[24]{\frac{1}{25}}$$

$$f(x) \text{ at } x = \sqrt[24]{\frac{1}{25}} = \left(\sqrt[24]{\frac{1}{25}}\right)^{25} = \frac{1}{(25)^{\frac{25}{24}}}$$

$$\therefore \text{The point } (u, f(u)) = \left(\sqrt[24]{\frac{1}{25}}, \frac{1}{25^{\frac{25}{24}}}\right)$$

Q2: Is $f(x) = \sin x$ uniformly continuous on $\mathbb{R}$?
Your answer with justification:

[3]

→ Yes, $f(x) = \sin x$ is uniformly continuous on $\mathbb{R}$

→ We know that $\sin x$ is continuous on $\mathbb{R}$ (as its Radius of convergence is ∞)

∴ $\sin x$ restricted to $[0, \frac{4\pi}{2}]$ is uniformly continuous
(As any continuous function in a closed interval is uniformly continuous)

→ We also know that $\sin x$ is periodic, with period 2π

∴ $\sin x = \sin x'$, for some $x' \in [0, 2\pi]$,

and similarly, $\sin y = \sin y'$, for $y' \in [0, 2\pi]$

→ As $\sin x$, $x \in [0, 4\pi]$ is uniformly continuous, Given an $\epsilon > 0$,
we can get a $\delta > 0$ such that $|f(x') - f(y')| < \epsilon$, when $|x' - y'| < \delta$,
 $x', y' \in [0, 4\pi]$

Let $\delta = \min \{\delta', \pi\}$

Without

~~loss of generality~~, assume $x < y$, will show if $|x - y| < \delta$, then $|f(x) - f(y)| < \epsilon$

↳ As $\sin x$ is periodic, then $\sin x = \sin x'$, $x' \in [0, 2\pi]$

Let $x - y = h \rightarrow x = y + h$ ($h < \delta$)

∴ $\sin(y) = \sin(x+h) = \sin(x'+h) \rightarrow x'+h \in [0, 4\pi]$, as $h < \delta \leq \pi$

∴ $|f(x) - f(y)| = |f(x) - f(x'+h)| < \epsilon$, ~~as~~ as x' and $x'+h \in [0, 4\pi]$
and $|x' - (x'+h)| = |h| < \delta \leq \delta'$

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∴ $|f(x) - f(y)| < \epsilon$, if $x - y < \delta$, ~~and~~ and can do similar for $y > x$

∴ $\sin x$ is uniformly continuous on $\mathbb{R}$

Q13

$$\lim_{x \rightarrow 0^+} \frac{\log(\sin x)}{\log(e^x - 1)} \quad \text{equals} \quad [4]$$

your answer with justification:

→ ~~Let~~ let $f(x) = \log(\sin x)$ $\lim_{x \rightarrow 0^+} f(x) = 0$ $\lim_{x \rightarrow 0^+} g(x) = 0$
 $g(x) = \log(e^x - 1)$ Please Think Think

∴ Via L'Hopital's rule, $\lim_{x \rightarrow 0^+} \frac{f(x)}{g(x)} = \lim_{x \rightarrow 0^+} \frac{f'(x)}{g'(x)}$ $f'(x) = \frac{1}{\sin x} \cdot \cos x$
 $g'(x) = \frac{1}{e^x - 1} \cdot e^x$

$L = \lim_{x \rightarrow 0^+} \frac{\frac{\cos x}{\sin x}}{\frac{e^x}{e^x - 1}} = \frac{\cos x (e^x - 1)}{\sin x \cdot e^x} \rightarrow$ obvious it is again a $\frac{0}{0}$ situation

~~$\lim_{x \rightarrow 0^+} \frac{f(x)}{g(x)} = \lim_{x \rightarrow 0^+} \frac{f'(x)}{g'(x)}$~~

$L = \lim_{x \rightarrow 0^+} \frac{h(x)}{k(x)} = \lim_{x \rightarrow 0^+} \frac{h'(x)}{k'(x)}$

$h(x) = \cos x (e^x - 1)$
 $h'(x) = -\sin x (e^x - 1) + \cos x \cdot e^x$
 $k(x) = \sin x \cdot e^x$
 $k'(x) = \cos x \cdot e^x + \sin x \cdot e^x$

∴ $L = \lim_{x \rightarrow 0^+} \frac{[-\sin x (e^x - 1) + \cos x \cdot e^x]}{\cos x \cdot e^x + \sin x \cdot e^x} = 1$

$\lim_{x \rightarrow 0^+} \frac{\log(\sin x)}{\log(e^x - 1)} = 1$